

Mathematics II (part 1)

PROBABILITY THEORY



Ryan Fraser Kirwan

- Three common **interpretations** of probability
- **Probabilities** of events
- **Conditional** probabilities
- Mutually exclusive events and **Independent** events
- Multiplicative rules of probabilities
- **Bayes' theorem**

Probability Theory

Our world is full of **uncertainties** with random outcomes that are unpredictable ...



Flipping coin



Tossing dice



Spinning wheel

In an attempt to make sense of these **uncertainties**, the **theory of probability** is discovered/invented.

PROBABILITY THEORY

INTERPRETATION OF PROBABILITY

Interpretation of probability

Definition of probability

Mathematically, probability is defined as a number between 0 to 1 (inclusive) that is assigned to a random outcome of uncertainties.

But how to assign it?

There are **two** instructive ways, called interpretations of probability:

- Classical (theoretical) probability
- Relative frequency (Empirical) probability



Interpretation of probability

Classical or objective probability

- In classical interpretation of probability, all random outcomes are considered to be **equally likely**.
- **Definition:** The probability of a **random event A** occurring is:
 - $P(A) = \frac{\text{(Number of different outcomes in A)}}{\text{(Total number of possible outcomes)}}$

Interpretation of probability

Classical or objective probability: E.G., 1

What is the
probability of
picking 6 of hearts ?

$1/5$



Interpretation of probability

Classical or objective probability: E.G., 2

What is the
probability of
picking a card with
hearts?

$2/5$



Interpretation of probability

Classical or objective probability: E.G., 3

What is the
probability of
picking a value
 ≥ 7 ?

$4/5$



Interpretation of probability

Classical or objective probability

- In classical interpretation of probability, all random outcomes are considered to be **equally likely**.
- **Definition:** The probability of a **random event A** occurring is:

The **random events** in the previous example were:

- (a) picking a 6,
- (b) picking a heart card and
- (c) picking a number ≥ 7

$$P(A) = \frac{\text{(Number of different outcomes in A)}}{\text{(Total number of possible outcomes)}}$$

This definition is not very practical for everyday life because it assumes the outcomes are equally likely.

Interpretation of probability

Relative Frequency Or Empirical Probability

- In **relative frequency** interpretation, the probability assigned to an outcome is the proportion of times it would occur over the long run.
- Such definition relies on **collecting data** over the years or existing past data to propose such relative frequencies.

$$P(A) = \frac{f}{n}$$

where f : number of times the outcomes occur,
 n : total number of observations.

Interpretation of probability

Relative Frequency Or Empirical Probability

Possible Concerns:

- How many times are considered long run?

- What if the outcome cannot be repeated?

E.g. probability of you getting an 'A' in this module?

- There is no past data as it is an utterly new situation?

For example, when Apple came out with the iPad, they didn't have past data on iPad sales.

$$P(A) = \frac{f}{n}$$

where f : number of times the outcomes occur,
 n : total number of observations.

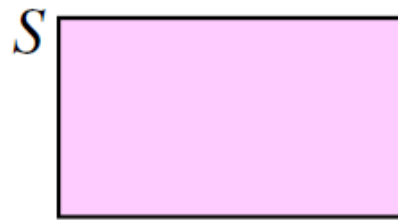
PROBABILITY THEORY

PROBABILITY OF EVENTS

Probability of Events

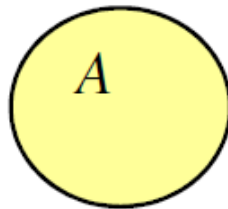
Definition and Rules

Fortunately, regardless of which interpretations of probability we use, the theory of probability is the same and applies to ALL.



Sample space S is **defined** as the **set** of **all possible outcomes** of a random phenomenon, typically drawn using a **Venn diagram** with a rectangle.

E.g.: sample space when tossing a dice, $S = \{1, 2, 3, 4, 5, 6\}$



Event A is **defined** as the set of **outcomes** that we are interested, typically drawn using a circle or ellipse inside the rectangle.

E.g.: getting a '6' when tossing a dice, $A = \{6\}$

Probability of Events

Definition and Rules: Example

E.g.: Let A denote the simple event of obtaining a '6' when tossing a dice,

$$P(A) = \frac{1}{6} \quad \text{if using classical interpretation}$$



$$P(A) = \frac{30}{100} \quad \text{if using relative frequency interpretation and you observed that this is a biased dice}$$

$$P(A) = 1 \quad \text{if using personal interpretation and assuming you were a magician who could control the outcome!}$$

Note: $P(A)$ is a notation to indicate probability of event A happening.

By **definition** of **probability**, the following **MUST** hold:

$$0 \leq P(A) \leq 1$$

and

$$P(S) = 1$$

Probability of Events

Definition and Rules : Complementary Events

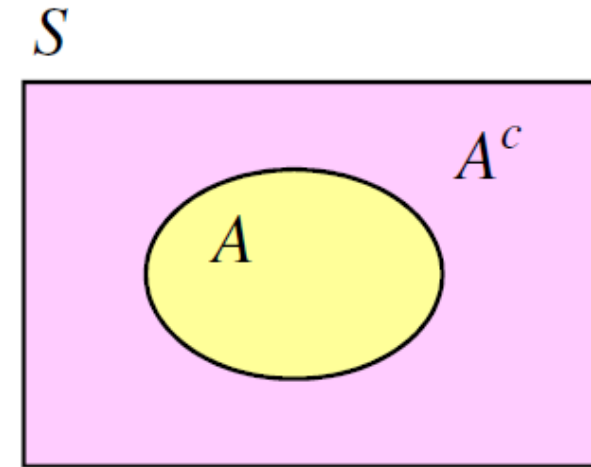
Probability of a complementary event is the probability that it doesn't happen.

Definition: The probability of a complementary event A^c is

$$P(A^c) = 1 - P(A)$$

Example:

- Tossing a dice, sample space $S = \{1, 2, 3, 4, 5, 6\}$
- Event A getting a '6' $A = \{6\}$



Probability of NOT getting a '6':

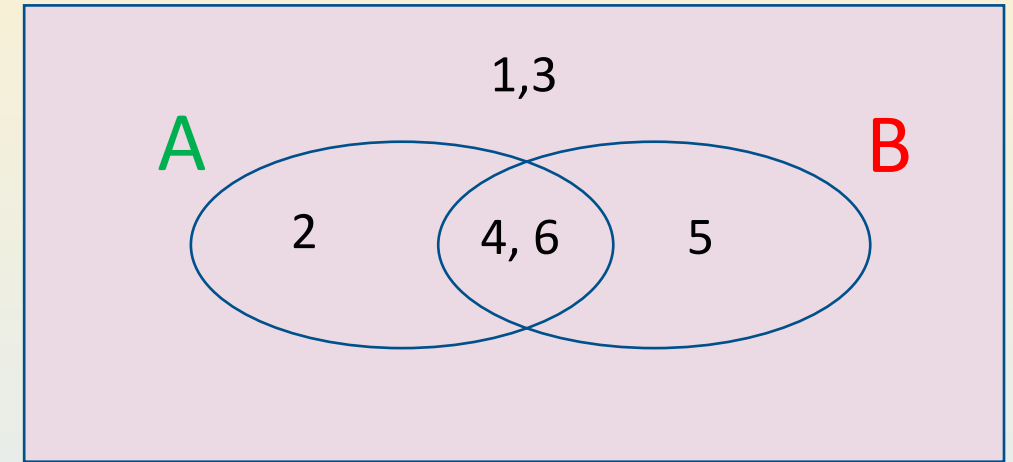
$$P(A^c) = 1 - P(A) = 1 - \frac{1}{6} = \frac{5}{6} \quad \text{if using classical interpretation}$$

$$P(A^c) = 1 - P(A) = 1 - \frac{30}{100} = \frac{70}{100} \quad \text{if using relative frequency interpretation (refer previous slide)}$$

Probability of Events

Dice Examples

- *Sample space:* $S = \{1, 2, 3, 4, 5, 6\}$
- **Event A** is getting an even number
 $A = \{2, 4, 6\}$
- **Event B** is getting a number > 3
 $B = \{4, 5, 6\}$



I want to know the probability of getting **an even number OR** a number > 3

What is the answer?

Consider this Venn diagram to aid in your reasoning.

Probability of Events

Dice Examples

- *Sample space:* $S = \{1, 2, 3, 4, 5, 6\}$

Probability of getting **an even number**
OR a number **> 3**

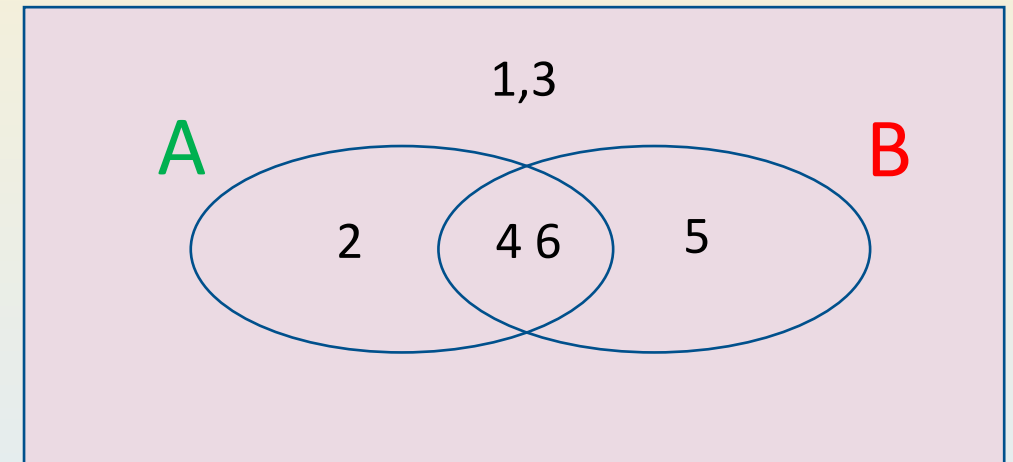
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{3}{6} - \frac{1}{3} = \frac{2}{3}$$

using **classical** interpretation

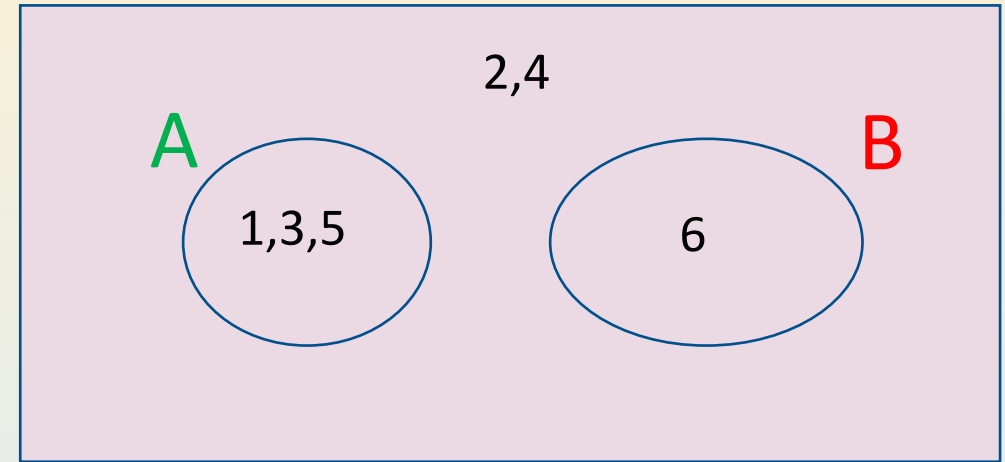
4/6



Probability of Events

Dice Examples

- *Sample space: $S = \{1, 2, 3, 4, 5, 6\}$*
- **Event A** is getting an odd number
- $A = \{1, 3, 5\}$
- **Event B** is getting a '6'
 $B = \{6\}$



Probability of getting an odd number or a '6' ?
(A OR B)

What is the answer?

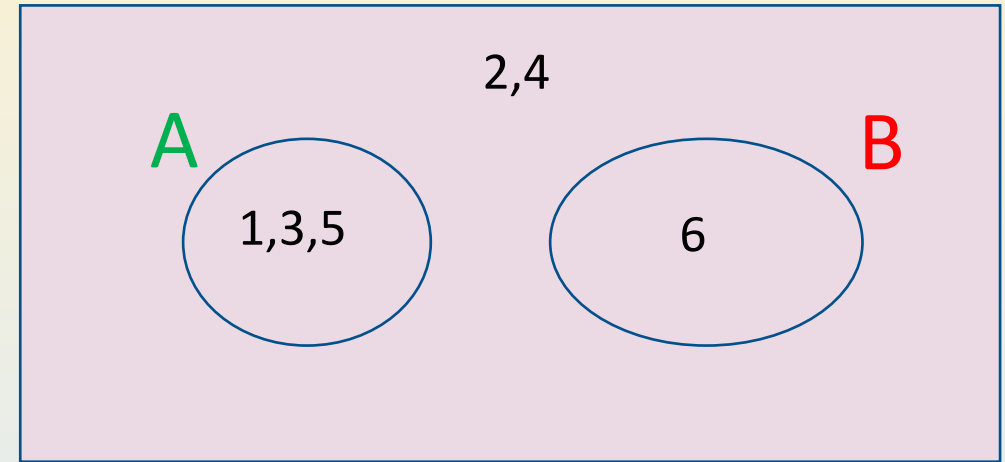
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Probability of Events

Dice Examples

- Sample space: $S = \{1, 2, 3, 4, 5, 6\}$
- Event A is getting an odd number
- $A = \{1, 3, 5\}$
- Event B is getting a '6'
 $B = \{6\}$

4/6



A and B are *Mutually exclusive events*

Probability of getting an odd number or a '6' ?
(A OR B)

What is the answer?

$$P(A \cup B) = P(A) + P(B) = \frac{3}{6} + \frac{1}{6} = \frac{2}{3}$$

Definition: For mutually exclusive events A and B :

$$P(A \cap B) = 0$$
$$\therefore P(A \cup B) = P(A) + P(B)$$

Probability of Events

Union and Intersection of Events

- The probability of **intersection** of events A and B is the probability that both A and B occur.
- The probability of **union** of events A and B is the probability of **either A or B or both** occur.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually exclusive events

$$\begin{aligned} P(A \cap B) &= 0 \\ \therefore P(A \cup B) &= P(A) + P(B) \end{aligned}$$

Probability of Events

Union and Intersection of Events: Example 1

A company has made a policy that in the next 5 years, 40% of their new employees will be male and 30% of their new employees will be Singapore-born. There will also be 35% overseas-born, female new employees. What percentage of new employees will be overseas-born?

Probability of Events

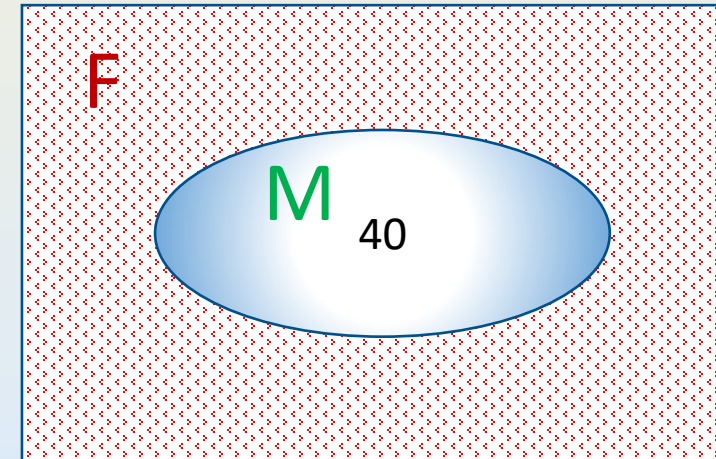
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Sample space: $S = \{\text{all new employees in the company}\}$

Event $M = \{\text{male employee}\} = 40$

Event $F = \{\text{female employee}\}$ (Complement of M) = 60



Probability of Events

Union and Intersection of Events: Example 2

A company has made a policy that in the next 5 years, 40% of their new employees will be male and 30% of their new employees will be Singapore-born. There will also be 35% overseas-born, female new employees. **What percentage of new employees will be Singaporean-born men?**

Sample space: $S = \{\text{all new employees in the company}\}$

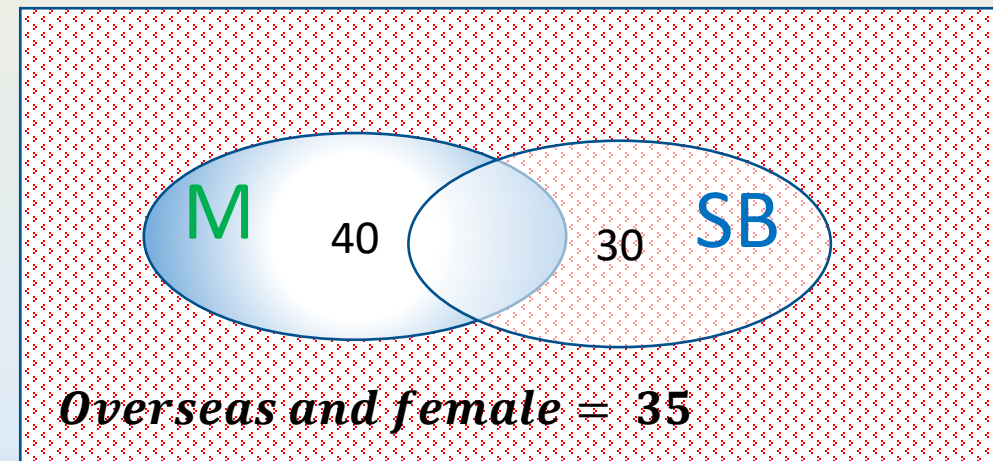
Event $M = \{\text{male employee}\} = 40$

Event $F = \{\text{female employee}\}$ (Complement of M) = 60

Event $SB = \{\text{Singaporean-born employee}\} = 30$

Event $O = \{\text{Overseas employee}\}$ (Complement of SB) = 70

$$p(O \cap F) = \text{Overseas and Female} = 35/100$$



Probability of Events

Union and Intersection of Events: Example 2

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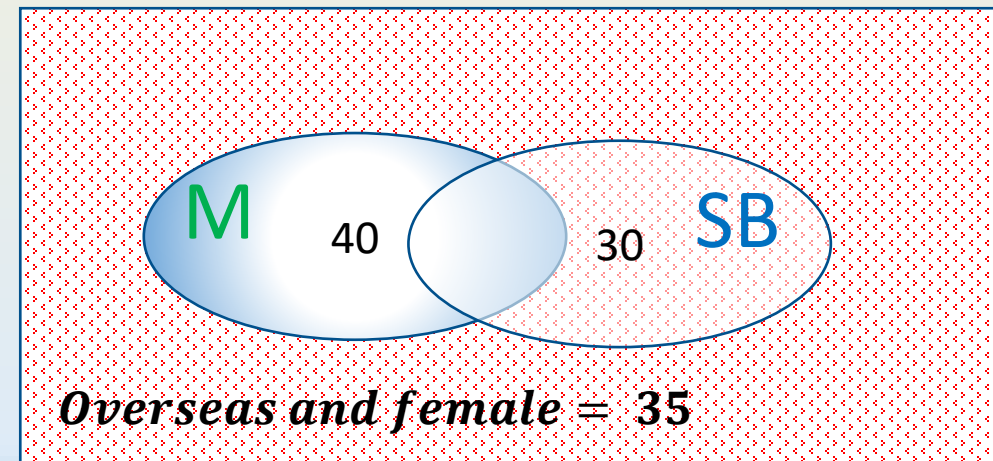
Event $O = \{\text{Overseas employee}\}$ (Complement of SB) = 70

$p(O \cap F) = \text{Overseas and Female} = 35/100$

Complement of this $p(O \cap F)$ is singapore-born or men

$$p(SB \cup M) = 1 - \left(\frac{35}{100}\right) = \frac{65}{100}$$

$$p(SB \cup M) = p(SB) + p(M) - p(SB \cap M)$$



Probability of Events

Union and Intersection of Events: Example 2

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$p(O \cap F) = \text{Overseas and Female} = 35/100$

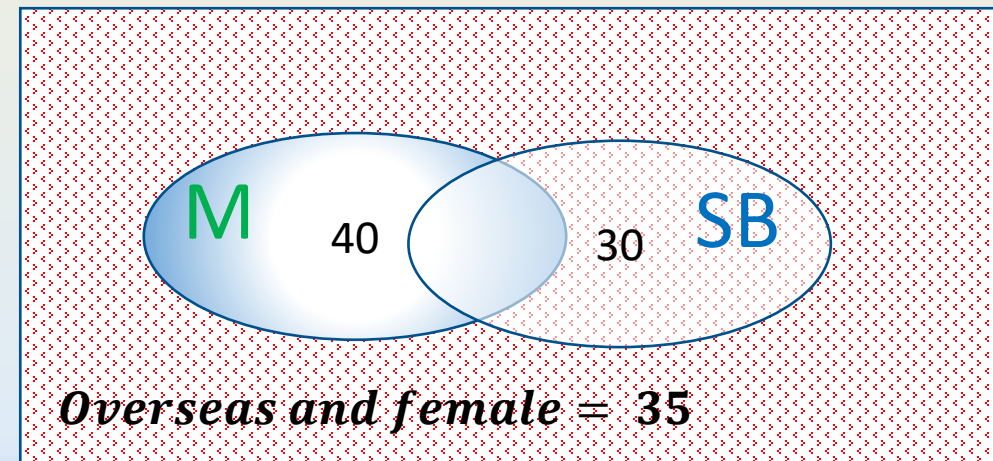
Complement of this $p(O \cap F)$ is singapore-born or men

$$p(SB \cup M) = 1 - \left(\frac{35}{100}\right) = \frac{65}{100}$$

$$p(SB \cup M) = p(SB) + p(M) - p(SB \cap M)$$

$$65 = 30 + 40 - p(SB \cap M)$$

$$p(SB \cap M) = 70 - 65 = 5$$



Probability of Events

Union and Intersection of Events: Example Alternative approach

- A company has made a policy that in the next 5 years, 40% of their new employees will be male and 30% of their new employees will be Singapore-born. There will also be 35% overseas-born, female new employees. **Fill the following table (one minute)**

	Men	Women	Total
Overseas	a	b	c
Singapore	d	e	f
Total	g	h	i

Probability of Events

Union and Intersection of Events: Example Alternative approach

Let M represents men, $P(M)=0.4$,

W represents women,

SB represents Singapore-born = $P(SB) = 0.3$

SM represents Singapore-born men = $P(SB \cap M)$?

SW represents Singapore-born women,

OW represents overseas-born women.

	Men	Women	Total
Overseas	0.35	0.35	0.7
Singapore	0.05	0.25	0.3
Total	0.4	0.6	1

Since M and W are complementary and mutually exclusive,

$$P(W) = 1 - P(M) = 1 - 0.4 = 0.6$$

Since SW and OW are mutually exclusive,

$$P(SW) = P(W) - P(OW) = 0.6 - 0.35 = 0.25$$

Similarly,

$$P(SM) = P(S) - P(SW) = 0.3 - 0.25 = 0.05$$

PROBABILITY THEORY

CONDITIONAL PROBABILITIES

Conditional probabilities

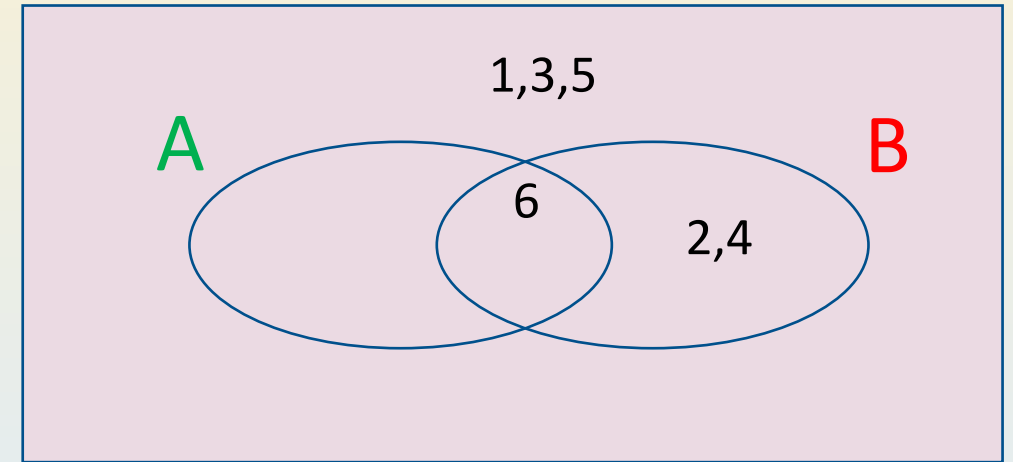
Introduction to Conditional Probabilities

- In practice, it happens naturally that **probability of an event A** can be **changed if additional information is available**, which leads to the discovery/creation of **conditional probability**.

- Tossing a dice, sample space $S = \{1, 2, 3, 4, 5, 6\}$
- Event A is getting a '6' = $1/6$
- Event B is getting an even number $B = \{2, 4, 6\} = 3/6$

Now, What is probability of event A Given that event B has occurred $P(A|B)$?

- Consider the Venn Diagram

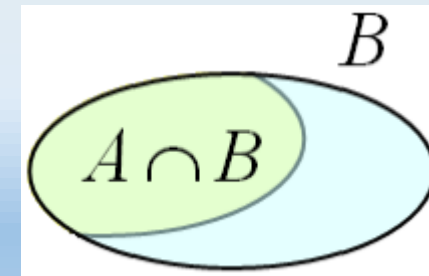
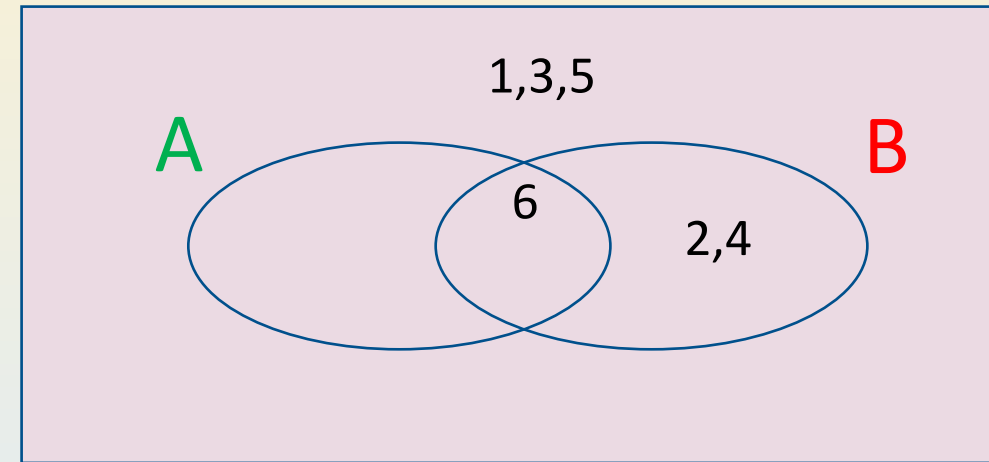


Conditional probabilities

Introduction to Conditional Probabilities

- Tossing a dice, sample space $S = \{1, 2, 3, 4, 5, 6\}$
- Event A is getting a '6' = $1/6$
- Event B is getting an even number $B = \{2, 4, 6\} = 3/6$
- What is probability of event A given that event B has occurred $P(A|B)$?
- **The sample space changed, it is not all S anymore,** it is only.
- From Venn Diagram:

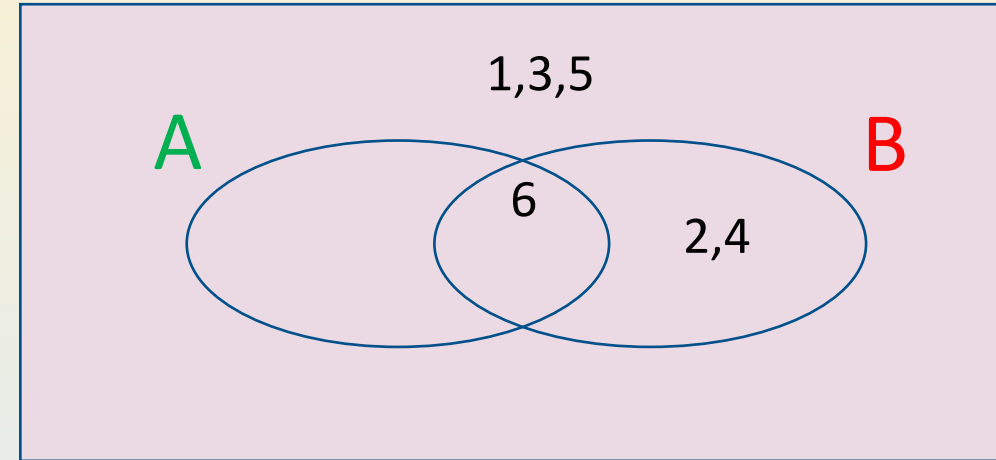
$$P(A|B) = \frac{1}{3} = \frac{\frac{1}{6}}{\frac{3}{6}}$$



Conditional probabilities

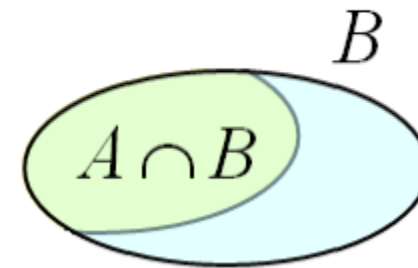
Conditional Probability Rule

$$P(A|B) = \frac{1}{3} = \frac{\frac{1}{6}}{\frac{3}{6}}$$



This leads us to the **definition** of **conditional probability** of event A given that event B has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Conditional probabilities

Conditional Probability Rule

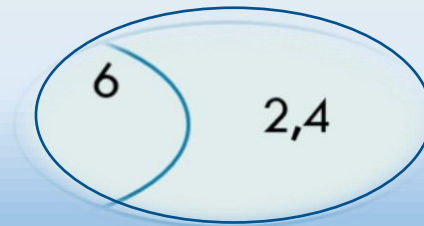
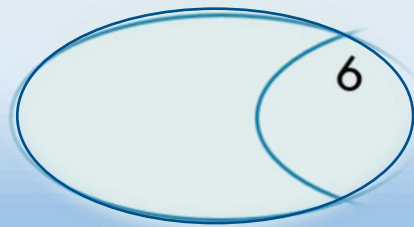
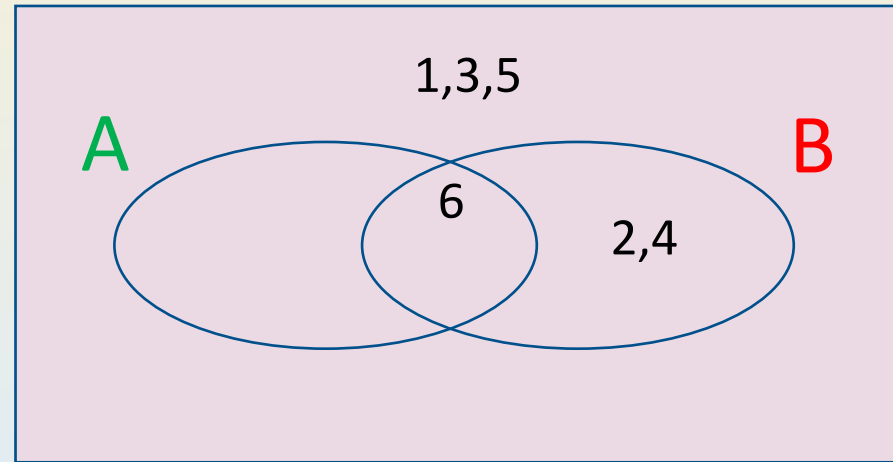
Is $P(B|A) = P(A|B)$?

Conditional probabilities

Conditional Probability Rule

Is $P(B|A) = P(A|B)$?

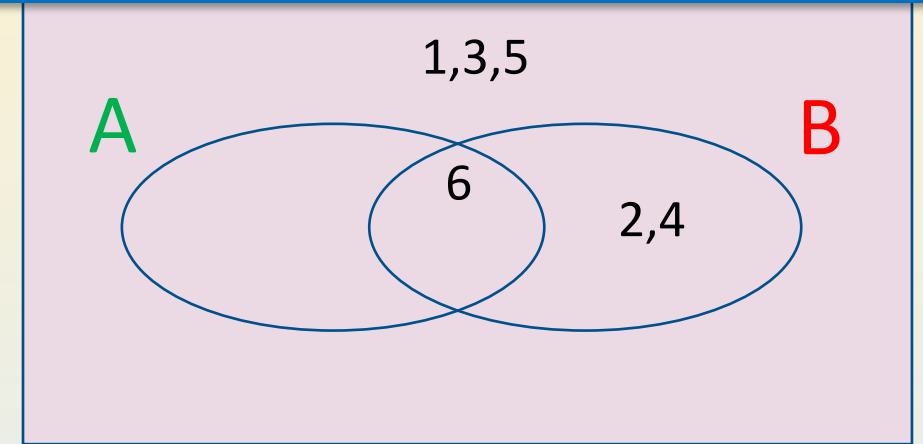
The **conditional probabilities** $P(A|B)$ and $P(B|A)$ represent **different probabilities** and are **NOT** the same!



Conditional probabilities

Conditional Probability Rule

- Intuitively, and from the diagram, what is probability of getting an even number given that a '6' has occurred ? I.E $P(B|A)$?



Using **definition** of **conditional probability**,

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{\frac{1}{6}}{\frac{1}{6}} = 1 \quad (\text{agree with intuition})$$
$$\neq P(A|B)$$

$$\begin{aligned} P(B \cup A) &= P(B) + P(A) - P(B \cap A) \\ P(B \cap A) &= P(B) + P(A) - P(B \cup A) \\ P(B \cap A) &= 1/2 + 1/6 - 1/2 \end{aligned}$$

Conditional probabilities

Conditional Probability Rule: Example (a)

The table below shows the probabilities of males (M) and females (F) being employed (E) or unemployed (U) in a population.

(a) Find the probability of being employed **Given** that the person is male: $P(E|M)$?

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

	M	F	Total
E	0.52	0.41	0.93
U	0.05	0.02	0.07
Total	0.57	0.43	1.00

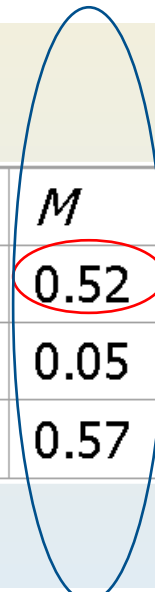
Conditional probabilities

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	M	F	Total
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Total	0.57	0.43	1.00

Probability of being **employed** **given** that the person is **male**

$$P(E|M) = \frac{P(E \cap M)}{P(M)} = \frac{0.52}{0.57} = 0.91$$

Conditional probabilities

Conditional Probability Rule: Example (b)

The table below shows the probabilities of males (M) and females (F) being employed (E) or unemployed (U) in a population.

(b) Find the probability of being male
Given that the person is employed:
 $P(M|E)$?

	M	F	Total
E	0.52	0.41	0.93
U	0.05	0.02	0.07
Total	0.57	0.43	1.00

Conditional probabilities

Conditional Probability Rule: Example (b)

The table below shows the probabilities of males (M) and females (F) being employed (E) or unemployed (U) in a population.

(b) Find the probability of being male
Given that the person is employed:
 $P(M|E)$?

	M	F	Total
E	0.52	0.41	0.93
U	0.05	0.02	0.07
Total	0.57	0.43	1.00

Probability of being **male given** that the person is **employed**

$$P(M | E) = \frac{P(M \cap E)}{P(E)} = \frac{0.52}{0.93} = 0.56$$

Conditional probabilities

Conditional Probability Rule: Summary

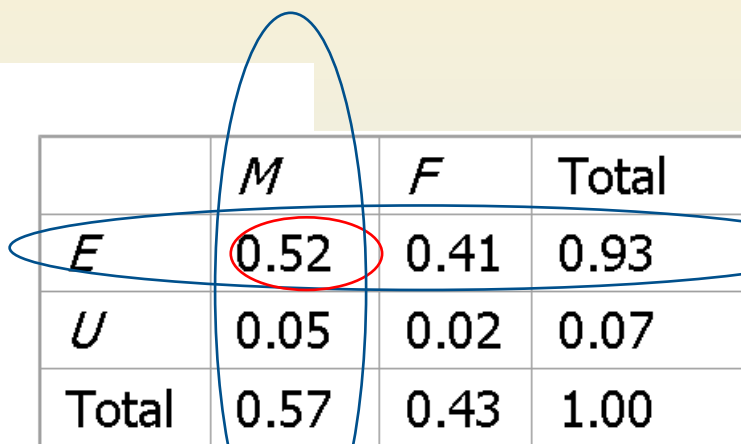
- (a) Probability of being **employed** given that the person is **male**

$$P(E|M) = \frac{P(E \cap M)}{P(M)} = \frac{0.52}{0.57} = 0.91$$

- (b) Probability of being **male** given that the person is **employed**

$$P(M|E) = \frac{P(M \cap E)}{P(E)} = \frac{0.52}{0.93} = 0.56$$

Note that: $P(E|M) \neq P(M|E)$



	<i>M</i>	<i>F</i>	Total
<i>E</i>	0.52	0.41	0.93
<i>U</i>	0.05	0.02	0.07
Total	0.57	0.43	1.00

PROBABILITY THEORY

MULTIPLICATIVE RULES OF PROBABILITIES

Multiplicative rules of probabilities

Introduction

The definition of **conditional probability** has given us a way to calculate the intersection of **events A and B**.

Since

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{and} \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

We have:

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B) = P(B|A)P(A)$$

These are known as the general **multiplication rules** of **probability**.

Multiplicative rules of probabilities

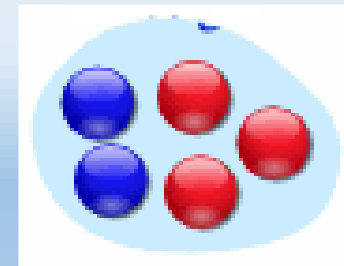
Solving Conditional Probability with Multiplication rules and Tree Diagrams (a)

The use of the **multiplicative rules** of probability together with a **tree diagram** is useful for solving problems involving **conditional probability**.

Example: **2 blue** and **3 red** marbles are in a bag. A child is asked to randomly pick 2 marbles **without replacement**.

- a) What is the probability that the child picks **a second blue marble, given** that he/she has already picked a **blue** marble?

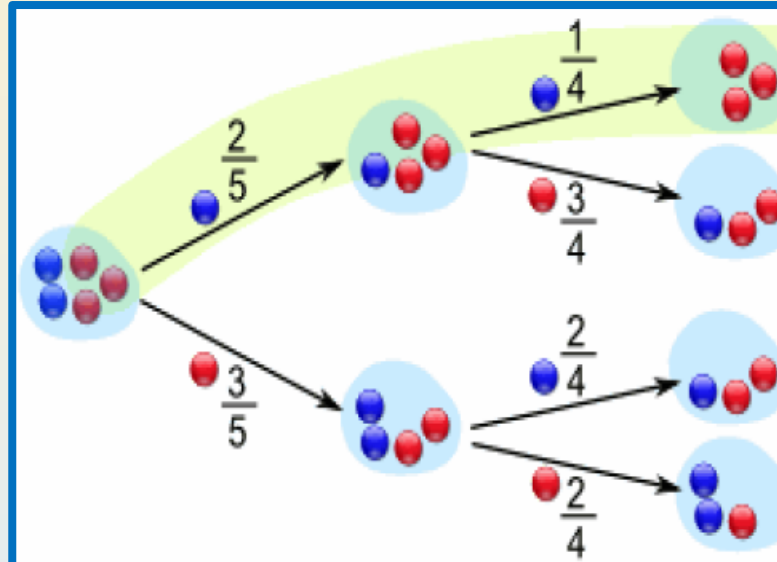
$$P(2^{nd} \text{ Blue} | 1^{st} \text{ Blue})$$



Multiplicative rules of probabilities

Solving Conditional Probability with Multiplication rules and Tree Diagrams (a)

Solution
(tree
diagram)



a. $P(2^{nd} B | 1^{st} B) = \frac{1}{4}$

Multiplicative rules of probabilities

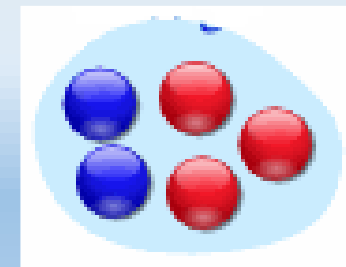
Solving Conditional Probability with Multiplication rules and Tree Diagrams (b)

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- a) What is the probability that the child picks **a second blue marble, given** that he/she has already picked a **blue** marble?
- b) What is the probability that the child picks **two blue marbles**?

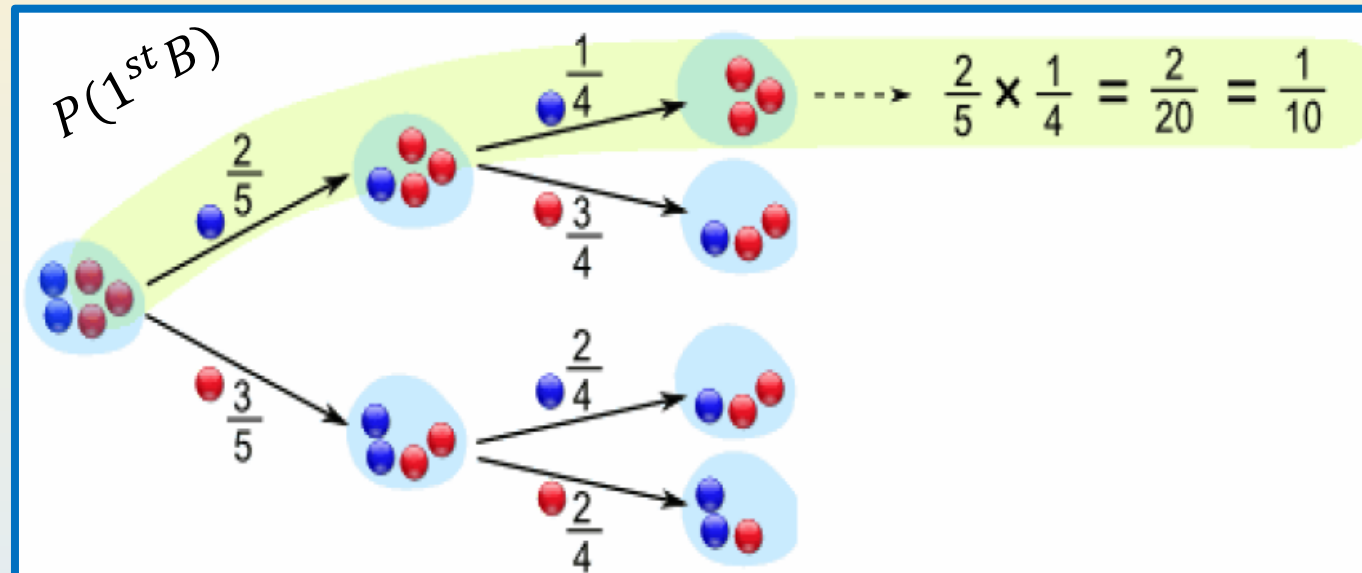
$$P(1^{st} B \cap 2^{nd} B)$$



Multiplicative rules of probabilities

Solving Conditional Probability with Multiplication rules and Tree Diagrams (b)

$$* P(2^{nd} B | 1^{st} B) = P(1^{st} B \cap 2^{nd} B)$$



a. $P(2^{nd} B | 1^{st} B) = \frac{1}{4}$

b. $P(2^{nd} B \cap 1^{st} B) = P(2^{nd} B | 1^{st} B)P(1^{st} B)$
 $= \left(\frac{1}{4}\right)\left(\frac{2}{5}\right) = \frac{1}{10}$

PROBABILITY THEORY

INDEPENDENT vs MUTUALLY EXCLUSIVE EVENTS

Independent vs Mutually Exclusive events

Independent Events?

If the **conditional probability** of A given that B has occurred **remains unchanged** as the **original probability** of A , then A and B are said to be independent events.

Definition: For independent events A and B

$$P(A|B) = P(A)$$

and

$$P(B|A) = P(B)$$

Example: consider tossing two fair dice

- Event A is getting a '6' on the first dice $P(A) = \frac{1}{6}$
- Event B is getting also a '6' on the second dice $P(B) = \frac{1}{6}$



Are events A and B independent or not?

Independent vs Mutually Exclusive events

Independent Events (yes)

If the **conditional probability** of **A** given that **B** has occurred **remains unchanged** as the **original probability of A**, then **A** and **B** are said to be independent events.

Definition: For independent events **A** and **B**

$$P(A|B) = P(A)$$

and

$$P(B|A) = P(B)$$

Example: consider tossing two fair dice

- Event **A** is getting a '6' on the first die $P(A) = \frac{1}{6}$
- Event **B** is getting also a '6' on the second die $P(B) = \frac{1}{6}$



Intuitively, probability of getting a '6' on the second die should not be influenced by the outcome of the first die, i.e. $P(B|A) = \frac{1}{6} = P(B)$

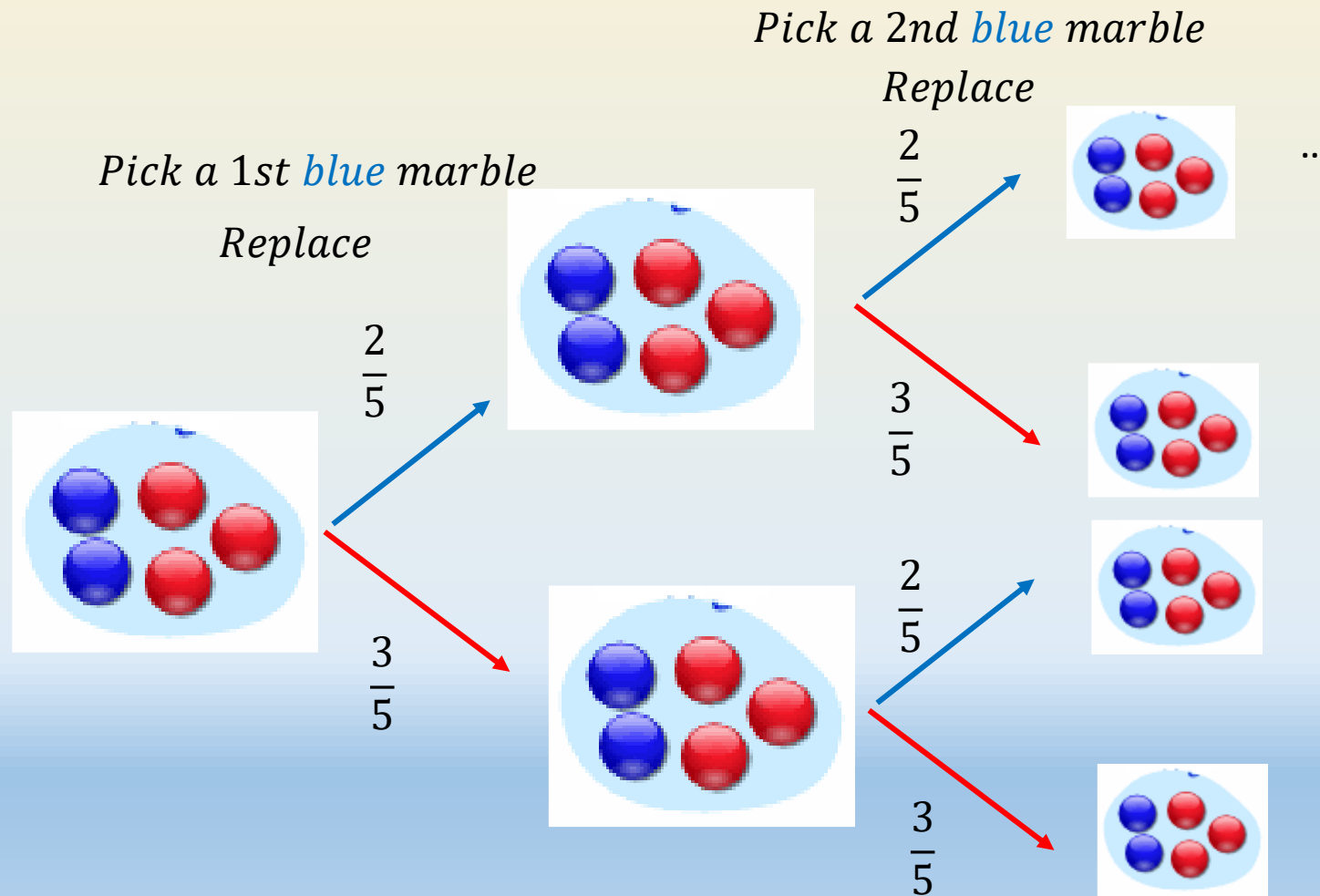
By **definition** of **conditional probability**,

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{\frac{1}{36}}{\frac{6}{36}} = \frac{1}{6} = P(B)$$

Independent vs Mutually Exclusive events

Independent Events: Marbles Example

Two blue marbles with replacements

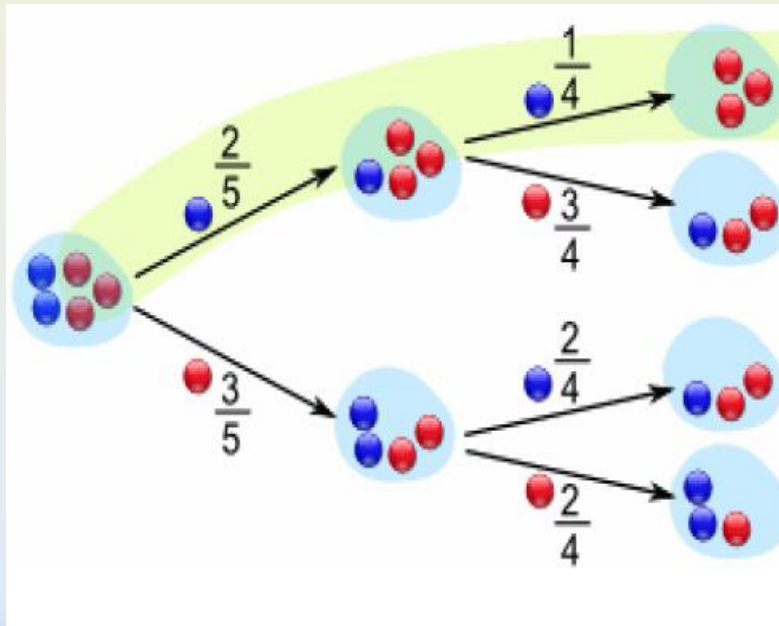


Independent vs Mutually Exclusive events

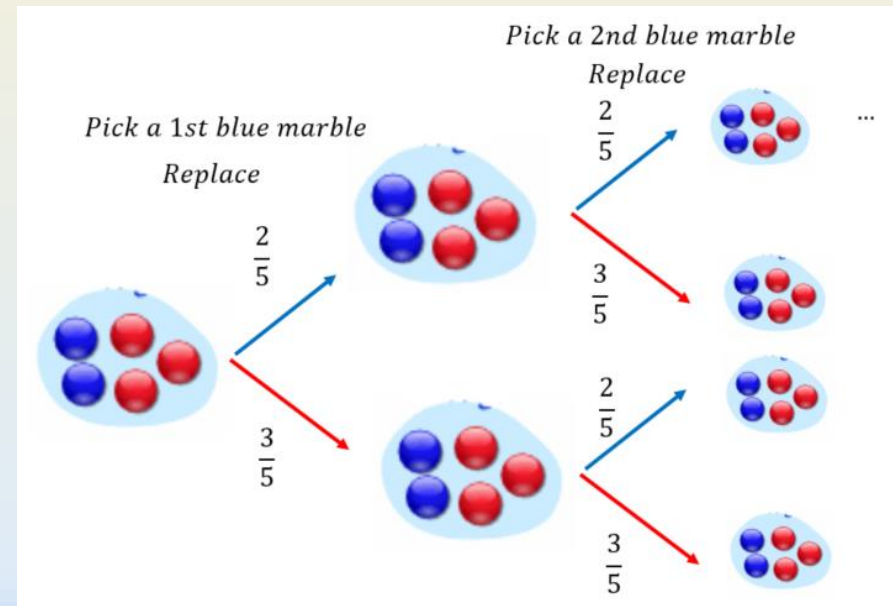
Independent Events: Marbles Example with/out replacements

Red and blue marbles

without replacement



with replacement



Independent vs Mutually Exclusive events

Intersection of Events

The definition of **independent events** has given us another way to calculate the **intersection of events** A and B .

For **independent events**,

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

We have:

$$P(A \cap B) = P(A|B)P(B) = P(A)P(B)$$

$$P(A \cap B) = P(B|A)P(A) = P(B)P(A)$$

These are known as the **multiplication rule** of **independent events**.

In fact, the **multiplication rules** can be extended to more than two independent events:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2)\dots P(A_n)$$

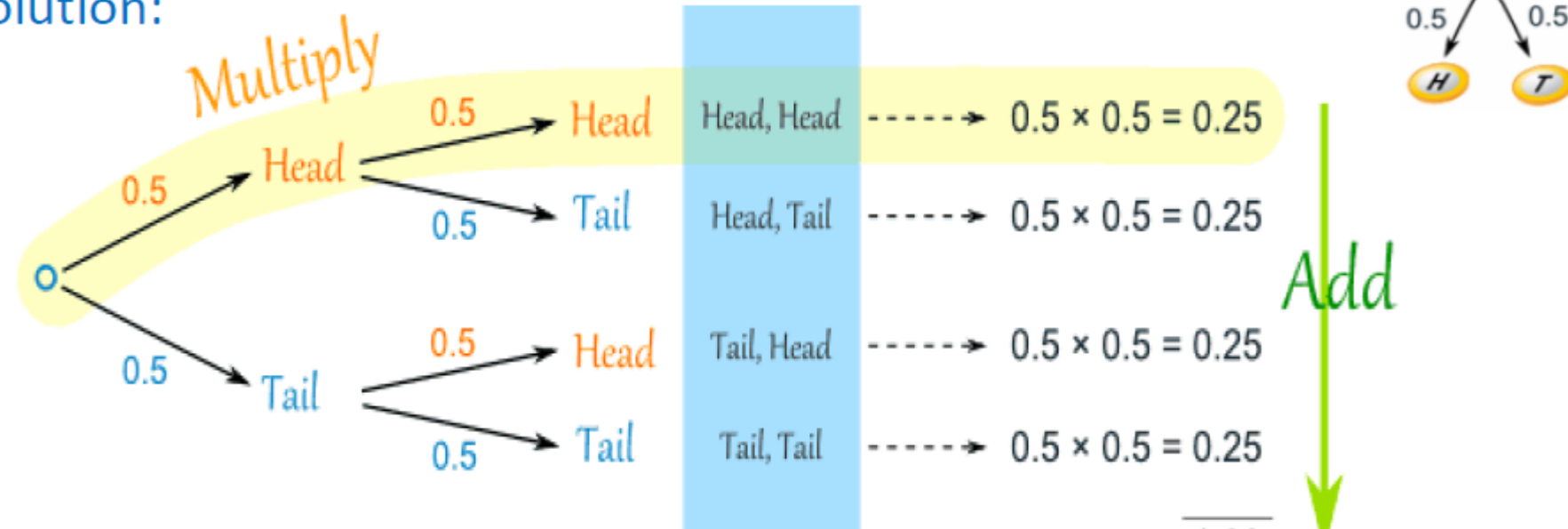
Independent vs Mutually Exclusive events

Intersection of Events: Example

Similarly, the use of the **multiplicative rules** together with a **tree diagram** is useful for solving problems involving **independent events**.

Example: Consider tossing two fair coins. What is the probability of getting at least one head?

Solution:



Independent vs Mutually Exclusive events

The Difference

- **Mutually exclusive** events will not occur at the same time (e.g. event of being male or female).
- **Independent events** will not affect the probabilities of one another (e.g. throwing two dices).

Independent vs Mutually Exclusive events

The Difference

- Mutually exclusive events will not occur at the same time.
- Independent events will not affect the probabilities of one another.
- Mutually exclusive events **MUST** be dependent because occurrence of one event is definitely affecting the other event.
- Independent events **CANNOT** be mutually exclusive because both events can occur together.

When Events Are:	$P(A \text{ or } B)$ is:	$P(A \text{ and } B)$ is:	$P(A B)$ is:
Mutually Exclusive	$P(A) + P(B)$	0	0
Independent	$P(A) + P(B) - P(A)P(B)$	$P(A)P(B)$	$P(A)$
Any	$P(A) + P(B) - P(A \text{ and } B)$	$P(A)P(B A)$	$\frac{P(A \text{ and } B)}{P(B)}$

PROBABILITY THEORY

BAYES' THEOREM + LAW OF TOTAL PROBABILITY

Bayes' theorem + law of total probability

Exhaustive Events

Two events A and B are called **exhaustive events** if they together cover all the possible outcomes.

Definition: For exhaustive events A and B

$$P(A \cup B) = P(S) = 1$$

Example:

- Tossing a dice, sample space
 $S = \{1, 2, 3, 4, 5, 6\}$
- Event A is getting an even number $A = \{2, 4, 6\}$
- Event B is getting an odd number $B = \{1, 3, 5\}$

Probability of getting an even number or odd number :

$$P(A \cup B) = P(S) = 1$$

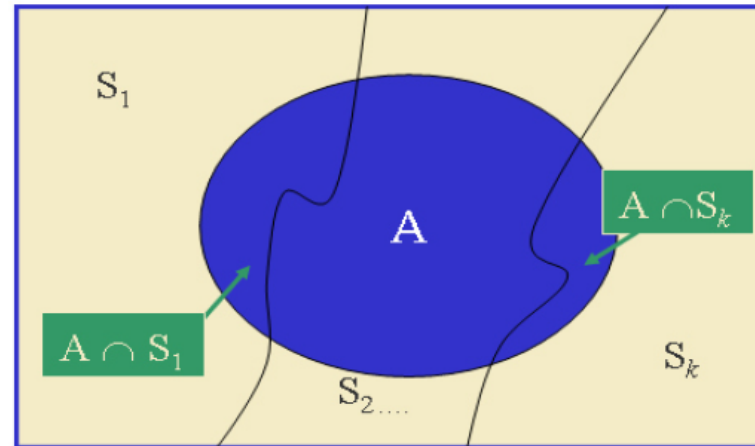
Bayes' theorem + law of total probability

Law of Total Probability

Given a set of **mutually exclusive** and **exhaustive events** ($S_1..S_k$), the probability of an event A ($P(A)$) can now be expressed using **law of total probability**.

Law of total probability:

$$P(A) = P(A \cap S_1) + P(A \cap S_2) + \dots + P(A \cap S_k)$$



$$P(A \cap B) = P(A|B)P(B)$$

Rewrite using **multiplicative rule of probability**,

$$P(A) = P(A|S_1)P(S_1) + P(A|S_2)P(S_2) + \dots + P(A|S_k)P(S_k)$$

Bayes' theorem + law of total probability

Bayes' theorem + Law of Total Probability

With the law of total probability, the conditional probabilities $P(A|B)$ and $P(B|A)$ can now be related by Bayes' theorem:

Since $P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(A \cap B)}{P(A)}$

we derive $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$

Rewrite $P(A)$ using law of total probability

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$

where B and B^c are mutually exclusive and exhaustive events

Bayes' theorem + law of total probability

Bayes' theorem in the real world: Example

Assume 1% of human population are allergic to cats. Suppose there is a medical test for this allergy with a correct detection rate of 80% and a false positive of 10%. What is the probability that when a person tests positive, they really have an allergy? **Should we invest in this medical TEST and spread its use?**

Bayes' theorem + law of total probability

Bayes' theorem in the real world: Example

Note that our world is full of uncertainties. Yes' may not always be 'Yes', and 'No' may not always be 'No', thus giving rise to the idea of false positive and false negative.

Assumptions

Truth

	They say you did	They say you didn't
You really did	<i>They are right!</i>	"False Negative"
You really didn't	"False Positive"	<i>They are right!</i>

Bayes' theorem + law of total probability

Bayes' theorem in the real world: Example

Assume 1% of human population are allergic to cats. Suppose there is a medical test for this allergy with a correct detection rate of 80% and a false positive of 10%. What is the probability that when a person tests positive, they really have an allergy?

Let: having Allergy $P(A)$ and not having $p(A')$ —*complement*

Let: test shows positive $P(T)$ and showing negative $P(T')$ —*complement*

$P(A) = 0.01$, $P(A') = 0.99$

$P(T|A) = 0.8$ (given that someone is allergic, the test is positive)

$P(T|A') = 0.1$ (given that someone is not allergic, the test is positive)

We want $P(A|T)$?

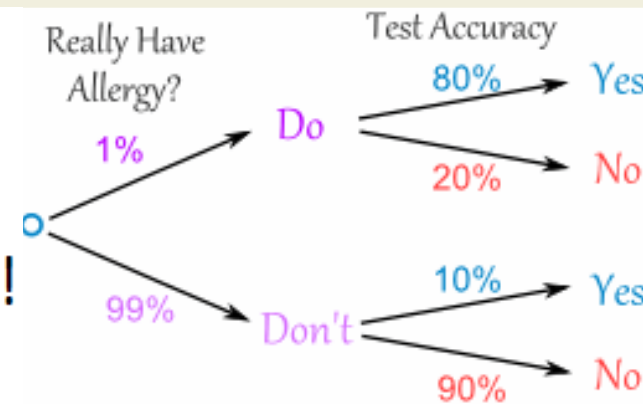
$$P(A|T) = \frac{P(T|A)P(A)}{P(T|A)P(A) + P(T|A^c)P(A^c)}$$

Bayes' theorem + law of total probability

Bayes' theorem in the real world: Example

- Let **A** be having the allergy, and **T** be being tested positive. Probability of really having the allergy given that test is positive is **$P(A|T)$** .

$$P(A|T) = \frac{P(T|A)P(A)}{P(T|A)P(A) + P(T|A^c)P(A^c)}$$
$$= \frac{(0.8)(0.01)}{(0.8)(0.01) + (0.1)(0.99)} = 0.074?? \text{only!!}$$



- At first glance, **$P(T|A)=0.8$** (accuracy) sounds quite good so you might be tempted to actually invest butif you do the math and get **$P(A|T)$** , it is only **0.074** useful considering the spread of the allergy in the population.

Bayes' theorem + law of total probability

Bayes' theorem in the real world: Monkey Example

Of the one million monkeys in Singapore most are good-natured. But 100 monkeys of the one million are pure evil! An aspiring student in our Course develops an “Evil Monkey Alarm” which he/she offers to sell to the government. The government decides to test the reliability of the alarm by conducting trials.

When presented with an **evil monkey**, the alarm goes off 99% of the time.

When presented with a **nice monkey**, the alarm goes off 1% of the time.



Bayes' theorem + law of total probability

Bayes' theorem in the real world: Monkey Example

(a) If a monkey sets off the alarm, what is the probability that it is evil?

We are given:

$$P(\text{nice}) = 0.9999, \quad P(\text{evil}) = 0.0001 \text{ (base rate)}$$

$$P(\text{alarm} | \text{nice}) = 0.01, \quad P(\text{alarm} | \text{evil}) = 0.99$$

$$P(\text{evil} | \text{alarm}) = \frac{P(\text{alarm} | \text{evil})P(\text{evil})}{P(\text{alarm})}$$

$$= \frac{P(\text{alarm} | \text{evil})P(\text{evil})}{P(\text{alarm} | \text{evil})P(\text{evil}) + P(\text{alarm} | \text{nice})P(\text{nice})}$$

$$= \frac{(0.99)(0.0001)}{(0.99)(0.0001) + (0.01)(0.9999)}$$

$$\approx 0.01$$

$P(\text{evil} | \text{alarm})$

We want to invert the conditional probability

Using Bayes Rule

Using law of total probability

To Sum up: Cheat Sheet

- Types of probabilities
- Probability Laws
- Sample Space, Random Event
- Union is OR
- Intersection/Joint is AND
- Mutually exclusive events have no intersection

$$P(A) = \frac{f}{n}$$

$$P(A) = \frac{|A|}{N}$$

$$0 \leq P(A) \leq 1$$

and

$$P(S) = 1$$

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\begin{aligned} P(A \cap B) &= 0 \\ \therefore P(A \cup B) &= P(A) + P(B) \end{aligned}$$

Probability Theory

Summary

Cheat Sheet

Definition of Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication Rules Of Probability to calculate intersection of events

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B) = P(B|A)P(A)$$

Dependent and independent events

$$P(A|B) = P(A)$$

and

$$P(B|A) = P(B)$$

Multiplication Rules for independent events

$$P(A \cap B) = P(A|B)P(B) = P(A)P(B)$$

$$P(A \cap B) = P(B|A)P(A) = P(B)P(A)$$

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n)$$

Probability Theory

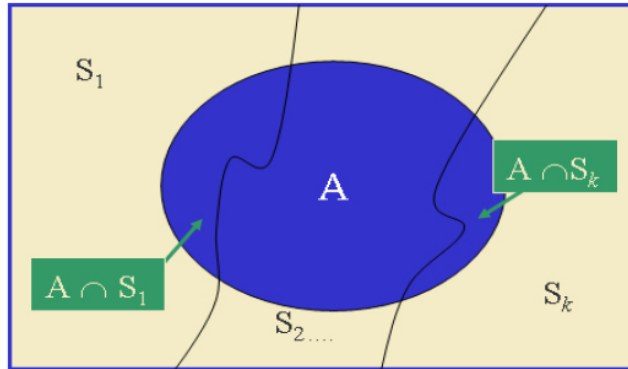
Summary

To Sum up (some more): Cheat Sheet

Law of total probability

Law of total probability:

$$P(A) = P(A \cap S_1) + P(A \cap S_2) + \dots + P(A \cap S_k)$$



Rewrite using multiplicative rule of probability,

$$P(A) = P(A|S_1)P(S_1) + P(A|S_2)P(S_2) + \dots + P(A|S_k)P(S_k)$$

$$P(A \cup B) = P(S) = 1$$

Bayes Theorem

Since $P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(A \cap B)}{P(A)}$

we derive
$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Rewrite $P(A)$ using law of total probability

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$